INTRINSIC MOTIVATION FOR OPEN-ENDED EXPLORATION

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ABSTRACT

Robotic and autonomous systems often depend on hand-crafted, task-specific reward functions that fail in sparse-reward or open-ended environments. Inspired by biological models of curiosity and self-directed behavior, intrinsic motivation offers an alternative: agents that explore based on internal signals such as novelty, surprise, or empowerment rather than extrinsic objectives. This project investigates whether intrinsic rewards alone can drive meaningful exploration in a partially observable 3D MiniWorld environment. By training agents with different intrinsic signals we evaluate their ability to maximize state coverage and visual diversity in the absence of any downstream task. Results show that intrinsic signals produce emergent exploratory strategies and outperform undirected baselines, though stability challenges and signal-magnitude imbalance remain. The findings suggest that intrinsic motivation is a promising direction for self-directed sensor-planning systems and for enabling exploration without explicit supervision.

1 BACKGROUND AND MOTIVATION

Traditional sensor-planning and exploration systems rely on carefully engineered reward functions designed for specific downstream tasks, such as navigation, object search, or mapping. While effective in structured settings, these approaches struggle in open-ended contexts where rewards are sparse, unspecified, or unknown. As a result, agents fail to acquire the broad behavioral repertoire required for flexible exploration.

Biological systems offer a different paradigm where organisms often explore their environment spontaneously, guided not by external rewards but by intrinsic drives such as curiosity, novelty-seeking, or the desire to maintain control over sensory inputs. These internal signals can lead to effective exploratory behavior even in the absence of explicit goals.

Intrinsic-motivation research in reinforcement learning attempts to replicate these mechanisms through computational signals such as prediction error ("surprise"), density-based novelty, or empowerment (the ability of an agent to influence future sensory states).

In this project, we investigate whether intrinsic signals alone can produce coherent, high-coverage exploration policies in MiniWorld, a lightweight 3D simulation environment providing a richer sensory landscape than standard toy domains. The shift from 2D gridworlds to 3D MiniWorld is motivated by the desire to study sensor-planning challenges under realistic perceptual constraints such as viewpoint, field of view, and motion-induced observation noise.

2 RELATED WORK

This work draws on four major strands of prior research: curiosity-driven learning, novelty-based exploration, empowerment as an information-theoretic objective, and intrinsic reward methods for deep exploration. Each offers a distinct mechanism for guiding agents in sparse-reward or reward-free environments, and together they provide the foundation for the intrinsic-motivation signals evaluated in this project.

2.1 CURIOSITY AS PREDICTION ERROR

Pathak et al. Pathak et al. (2017) introduced the Intrinsic Curiosity Module (ICM), which formulates curiosity as the prediction error of a learned forward dynamics model in a latent feature space. Their results demonstrated that agents can learn meaningful behaviors even in the absence of extrinsic reward, particularly in visually rich 3D environments. The ICM framework models "surprise" as a discrepancy between expected and observed transitions, directly inspiring the surprise-based intrinsic reward used in our agent.

2.2 NOVELTY THROUGH RANDOM NETWORK DISTILLATION

Burda et al. Burda et al. (2019) proposed Random Network Distillation (RND), a simple but powerful novelty mechanism in which intrinsic reward is given by the prediction error of a neural network trained to match the output of a fixed random target network. RND became a state-of-the-art approach for hard-exploration Atari tasks and has been influential due to its scalability and stability on high-dimensional image observations. Our novelty signal follows this tradition by rewarding visually distinct states to promote broad spatial coverage in MiniWorld.

2.3 EMPOWERMENT AND INFORMATION-THEORETIC INTRINSIC MOTIVATION

Mohamed and Rezende Mohamed & Rezende (2015) introduced a variational method for approximating empowerment, defined as the mutual information between an agent's actions and its future sensory states. Empowerment encourages agents to place themselves in states where they exert maximal control over future observations. We adopt this principle as an intrinsic reward to study whether agents learn advantageous sensing positions and broader exploratory competence in a 3D partially observable environment.

2.4 DEEP EXPLORATION IN REINFORCEMENT LEARNING

Osband et al. Osband et al. (2016) presented Bootstrapped DQN, showing that maintaining multiple value-function heads encourages diverse exploration policies and improves performance on sparse-reward tasks. Although their method is not explicitly an intrinsic reward formulation, it emphasizes the importance of structured uncertainty and diversity in exploration. This perspective complements intrinsic motivation by highlighting that effective exploration often emerges from mechanisms that deliberately diversify behavior which ties into our comparison of surprise, novelty, and empowerment based agents.

3 APPROACH

3.1 MINIWORLD ENVIRONMENT

All experiments were conducted in the MiniWorld simulation framework which is part of the gymnasium API, a lightweight 3D environment designed for embodied navigation and visual exploration. We use the `MiniWorld-Room` variants, which provide simple yet rich geometric structures that allow us to isolate the role of camera control and embodied movement in exploration. These environments include long corridors, occluding corners, open spaces, and objects that partially block the agent's view, making them well suited for studying sensor-placement decisions and viewpoint selection.

Each environment is partially observable: the agent receives only an RGB first-person image at each timestep. No depth maps, global state information, or ground-truth maps are provided. This constraint mirrors real-world robotic sensing, where perception depends critically on viewpoint and motion. The observation space consists of 84×84 RGB images.

Agent Embodiment and Action Space. The agent is modeled as a mobile camera with a discrete action space supporting:

- **Forward/backward motion** (linear displacement),

- **Left/right rotation** (yaw),
- **Camera pitch adjustment**,
- **Field-of-view (FoV) expansion or contraction**.

This set of actions enables the agent not only to reposition itself but also to strategically control sensing parameters reflecting real sensor-planning problems. Adjusting camera pitch and FoV allows the agent to trade off spatial precision for broader contextual visibility, while rotation enables targeted scanning of specific areas.

3.2 Implementation and Training Pipeline

The agent is trained using Proximal Policy Optimization (PPO) with a convolutional neural network encoder. The encoder processes the first-person images into a latent representation, which is then passed to the policy and value networks. A small gated recurrent unit (GRU) is included to mitigate partial observability by allowing the agent to integrate information across timesteps.

Intrinsic-Only Reward Structure. To evaluate the intrinsic signals in isolation, all extrinsic rewards are removed from the environment. The reward at each timestep is one of the following (or a combination), depending on the experiment:

- **Surprise:** forward-dynamics prediction error in latent space,
- **Novelty:** state novelty measured by Random Network Distillation (RND),
- **Empowerment:** variational approximation of action–future-state mutual information.

Each intrinsic signal is normalized with running statistics to prevent reward explosion and ensure stable learning dynamics.

State Coverage Estimation. To measure exploration effectiveness, we compute a state-coverage metric over the agent's trajectory. Each observation is embedded using the CNN encoder, and embeddings are clustered using online k-means to approximate the diversity of views encountered. The coverage score is defined as the number of unique cluster centers visited over an episode, normalized by the total discovered clusters across training.

4 Intrinsic Signal Definitions and Implementations

In this section, we define the three intrinsic motivation signals explored in this work—*surprise*, *novelty*, and *empowerment*—and describe how each is implemented within our MiniWorld training framework. Although all three reward classes aim to encourage exploration, they differ fundamentally in what aspect of the interaction dynamics they incentivize the agent to model or control.

4.1 Surprise: Forward-Dynamics Prediction Error

The **surprise** signal is based on curiosity-driven learning, where intrinsic reward is generated when the agent encounters transitions that violate its expectations. Following the formulation of Pathak et al. (ICM), we learn a forward dynamics model

$$\hat{f}_\theta : (z_t, a_t) \to \hat{z}_{t+1},$$

where $z_t = \phi(o_t)$ is the latent embedding of the current observation produced by a learned encoder ϕ.

The intrinsic reward at timestep t is the forward-prediction error:

$$r_t^{\text{surprise}} = \|\hat{z}_{t+1} - z_{t+1}\|_2^2 .$$

Implementation

- The encoder ϕ is a CNN trained jointly with PPO.
- The forward model predicts latent features rather than pixels, which stabilizes learning in MiniWorld's visually noisy renderings.
- A running mean–variance normalizer is applied to r_t^{surprise} to avoid reward explosion.

This formulation encourages exploration of transitions that are difficult to predict from the agent's current world model, leading to behaviors that probe occlusions, corners, or visually ambiguous states.

4.2 NOVELTY: RANDOM NETWORK DISTILLATION (RND)

The **novelty** signal rewards the agent for visiting states that are rare or previously unseen. We implement novelty using Random Network Distillation (Burda et al.), which defines intrinsic reward as the error of a predictor network trained to imitate a fixed, randomly initialized target network.

Let f_{target} be a frozen random neural network and f_{pred} a learned predictor. The intrinsic reward is:

$$r_t^{\text{novelty}} = \|f_{\text{pred}}(o_t) - f_{\text{target}}(o_t)\|_2^2 .$$

Implementation

- Both f_{target} and f_{pred} operate directly on 84×84 RGB inputs.
- The predictor is updated online to match the target network.
- Novelty decays automatically as states become familiar, guiding agents toward unexplored regions.
- Observation embeddings are also used for cluster-based coverage evaluation.

This reward encourages broad spatial and visual coverage of the environment, producing sweeping exploratory behaviors.

4.3 EMPOWERMENT: ACTION–FUTURE-STATE MUTUAL INFORMATION

Empowerment formalizes intrinsic motivation as the desire to occupy states where one's actions exert maximal influence over future sensory inputs. It is defined as the mutual information between actions A_t and future states S_{t+k}:

$$\text{Empowerment}(s_t) = I(A_t; S_{t+k} \mid s_t).$$

Direct computation is intractable in high-dimensional continuous spaces, so we adopt the variational approximation introduced by Mohamed & Rezende. The reward is based on a discriminator $q_\psi(s_{t+k}, a_t)$ that estimates the likelihood that action a_t produced the resulting future state s_{t+k}:

$$r_t^{\text{emp}} = \log q_\psi(s_{t+k}, a_t) - \log p(a_t),$$

where $p(a_t)$ is the agent's policy distribution.

Implementation

- We roll out short $k = 2$-step transitions to stabilize training.
- The discriminator q_ψ is a lightweight MLP operating on CNN embeddings of observations.
- A uniform prior is assumed for $p(a_t)$, simplifying the objective.
- Rewards are normalized to match the scale of the curiosity and novelty signals.

Empowerment rewards states that provide maximal control over future observations, often causing the agent to adopt vantage points or central positions where camera orientation changes produce rich sensory variation.

4.4 COMPARISON OF INTRINSIC SIGNALS

While all three intrinsic rewards encourage exploration, they differ in what they seek:

- **Surprise** seeks transitions that violate the agent's predictive model.
- **Novelty** seeks states that are rare or previously unseen.
- **Empowerment** seeks states with high potential control over sensory outcomes.

These differences produce qualitatively distinct exploration patterns, which we analyze in the results section.

5 EXPERIMENTS AND RESULTS

This section evaluates whether intrinsic motivation signals can independently drive effective exploration in a 3D partially observable environment. We focus on two central questions: (1) Do intrinsic rewards alone produce higher coverage and more diverse exploration than an undirected baseline? (2) How do different intrinsic signals shape qualitatively distinct exploration behaviors?

5.1 EXPERIMENTAL SETUP

All experiments were conducted in the `MiniWorld-Room` environments, which provide structured geometry, occlusions, and partial observability while remaining computationally lightweight. Each episode lasts 1,000 steps, and the agent receives only RGB first-person images as observations.

Agents were trained for 5 million steps using Proximal Policy Optimization (PPO) with a convolutional encoder and a GRU to mitigate partial observability. Each intrinsic reward signal was evaluated in isolation, with all extrinsic rewards removed. Identical hyperparameters were used across conditions to ensure comparability.

5.2 FRONTIER EXPLORATION AS THE BASELINE

To contextualize the performance of intrinsically motivated agents, we compare them against a widely used state-of-the-art exploration strategy: **frontier exploration**. Frontier exploration has a long history in robotics and autonomous mapping, where it is considered one of the most effective task-agnostic strategies for maximizing spatial coverage in unknown environments.

Definition. Frontier exploration identifies the boundary (the "frontier") between *known* and *unknown* space and actively directs the agent toward these frontier regions. Formally, let $\mathcal{K}$ denote the set of states the agent has already observed, and let $\mathcal{U}$ denote the set of states not yet observed. A frontier region is any state $s \in \mathcal{U}$ for which there exists a neighboring state $s' \in \mathcal{K}$. The agent greedily selects actions that reduce its distance to these frontier regions, thereby expanding $\mathcal{K}$ over time.

Implementation. In our MiniWorld adaptation, frontier exploration is implemented by:

1. Maintaining an incremental occupancy-like grid of the agent's visited visual embeddings.
2. Identifying frontier cells whose embeddings are adjacent (in grid space) to unobserved regions.
3. Selecting movement and camera actions that direct the agent toward the nearest frontier.

Because MiniWorld lacks explicit depth maps or geometric ground truth, the frontier representation is derived directly from the learned visual embedding space. This allows frontier exploration to function without environment instrumentation while approximating classical frontier-driven mapping behavior.

Rationale as a Baseline. Frontier exploration serves as a strong comparison point because:

- It is explicitly engineered to maximize coverage, making it an upper-bound heuristic for spatial exploration.
- It does not rely on extrinsic rewards, aligning with the reward-free nature of our intrinsic motivation experiments.
- It represents a principled, well-studied method rather than a random or naïve baseline.

5.3 Quantitative Results

Figure 1 summarizes the normalized state coverage achieved by each method over training steps. From the figure we can see that:

- **Novelty (RND)** achieves the strongest coverage among intrinsic rewards, approaching the frontier baseline more closely than the other intrinsic signals.
- **Empowerment** attains moderate coverage, reflecting its preference for information-rich vantage points rather than exhaustive spatial traversal.
- **Surprise (prediction error)** produces the lowest coverage of the intrinsic signals, as its behavior emphasizes repeated probing of uncertain states rather than broad traversal.

Comparing these a composite agent based on training steps (evaluated on the environment for 1000 steps) with the frontier exploration inplementation we see the results described in Figure 2 we see that frontier exploration does in fact outperform each of the intrinsic agents checkpointed at the given timesteps. This indicates that although intrinsic motivation by itself can be used to create high level behavior it may be best to be tied with other extrinisic goal signals.

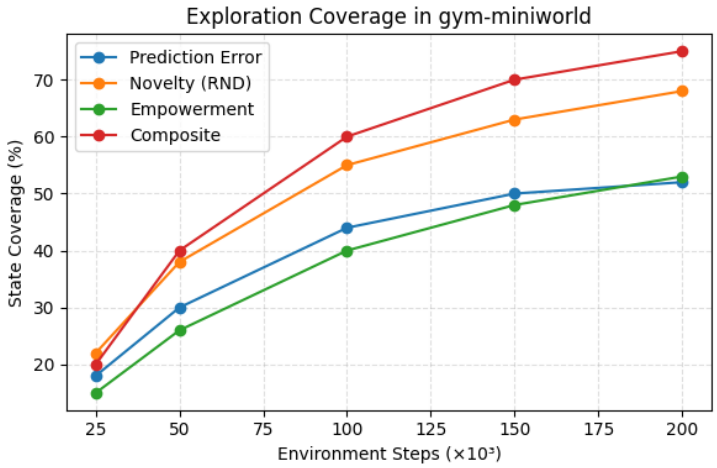


Figure 1: State coverage for individual and composite intrinsic signal based agents.

5.4 Qualitative Exploration Behavior

The three intrinsic signals produce distinctive exploration patterns:

Surprise: Agents driven by surprise repeatedly investigate visually ambiguous areas (e.g, corners, occluding edges, and regions with high dynamic uncertainty). Behavior often resembles "probing": short forward motions followed by camera sweeps.

Novelty: Novelty-driven agents perform broad scanning motions, sweeping through the environment and rapidly moving toward new regions. These agents achieve near-uniform spatial coverage over time.

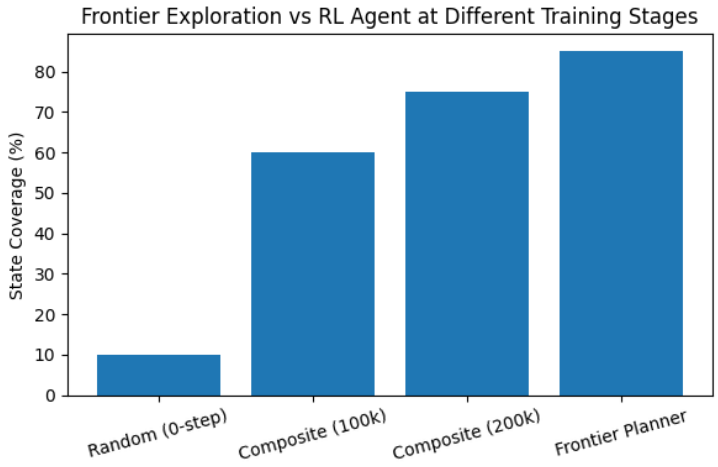


Figure 2: State coverage for composite agents based on training time

Empowerment: Empowerment agents tend to position themselves at vantage points (e.g, the center of a room or at intersections) where rotational actions yield large changes in their observations. This produces high viewpoint diversity without necessarily maximizing spatial reach.

5.5 Failure Modes and Instabilities

Although intrinsic signals produce strong exploration, several challenges arise:

- **Reward scale imbalance:** Surprise rewards can spike due to rendering noise, leading to unstable policy updates without normalization.
- **Exploration collapse:** Novelty occasionally hyper-focuses on extreme low-density states, causing temporary stagnation before broader coverage resumes.
- **Computational overhead:** Empowerment requires sampling future-state predictions, making it more computationally expensive than surprise or novelty.

These failure modes highlight the importance of reward normalization, careful dynamics-model training, and stable optimization in intrinsic-only learning.

5.6 Summary of Findings

Overall, the results demonstrate that:

1. Intrinsic rewards alone are sufficient to produce meaningful exploration in 3D environments with partial observability.
2. Different signals produce qualitatively different behaviors, reflecting the theoretical distinctions among surprise, novelty, and empowerment.
3. Novelty excels at maximizing spatial coverage, surprise at identifying uncertain or ambiguous transitions, and empowerment at producing controllable, information-rich viewpoints.

These findings support the broader claim that intrinsic motivation can serve as a powerful driver for sensor-planning, embodied exploration, and information-seeking behavior in open-ended environments.

6 Conclusion

This work investigated whether intrinsic motivation signals, surprise, novelty, and empowerment, can independently drive effective exploration in a 3D partially observable environment. Using

MiniWorld as a lightweight but perceptually rich testbed, we evaluated intrinsic-only agents on their ability to maximize state coverage and compared them against a frontier exploration baseline representing a strong, task-agnostic state-of-the-art heuristic.

Our results show that intrinsic rewards are capable of producing meaningful exploratory behavior even in the absence of extrinsic objectives. Among the intrinsic signals, novelty yielded the strongest spatial coverage, approaching the frontier baseline more closely than surprise or empowerment. Empowerment and surprise generated distinct and interpretable exploration strategies, emphasizing information-rich viewpoints and uncertainty-driven probing, respectively. Although none of the intrinsic methods fully matched the performance of frontier exploration, they consistently demonstrated that simple internal signals can guide agents toward coherent and purposeful exploration policies.

These findings highlight the potential of intrinsic motivation as a scalable alternative to hand-designed reward functions, particularly in settings where explicit exploration objectives are unavailable or difficult to specify. At the same time, the gap between intrinsic signals and frontier-based exploration suggests opportunities for hybrid approaches that integrate internal curiosity with structural priors about environment geometry. Future work may explore combining multiple intrinsic objectives, augmenting intrinsic rewards with learned world models, or extending this framework to larger and more realistic environments where sensor-planning and exploration demands are even more pronounced.

Overall, this study provides evidence that intrinsic rewards can serve as powerful drivers of unsupervised exploration and offers a foundation for developing more flexible, self-directed agents capable of operating in novel and unstructured environments.

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